Homework 7 — changed numbers (extra practice)

Module 7b - PK-PD Relationship · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A 70-kg volunteer received a single IV bolus dose of an investigational drug. The drug was administered at a dose of 12 mg/kg. The drug has an apparent volume of distribution that is 0.25 L/kg and an elimination half-life approximately 5 hours.

1a What is the AUC (area under the curve) following the IV bolus dose?

1b If the minimum effective concentration (MEC) is 4 mg/L, what is the duration of activity of this drug in this volunteer?

1c What is the duration of activity if the dose were 6 mg/kg?

1d What dose would provide a duration of action of 10 hours?

2. A 72-kg, 31-year-old female is receiving phenytoin for the treatment of seizures. When this patient was taking a daily oral dose of 250 mg, her steady state plasma concentration was 6.4 mg/L. The daily dose was increased to 350 mg which resulted in

a steady-state plasma concentration of 17.6 mg/L. Assume 100% bioavailability and an apparent volume of distribution that is equal to 70% of body weight.

2a What is this patient's KM?

2b What is the Vmax in this patient?

2c What is the daily dose rate of phenytoin that should achieve a steady state phenytoin plasma concentration of 14 mg/L in this patient? (Round to the nearest ten)

2d What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 14 mg/L? What is the half-life?

2d What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 14 mg/L? What is the clearance?

2e What is the expected steady state concentration if this patient were to take 300 mg/day?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: 346.3 (mg/L)·hr

WHAT THE STEM GIVES

weight	70-kg	scales the dose and the volume
D_0	12 mg/kg	× 70 kg = 840 mg
V_D	0.25 L/kg	× 70 kg = 17.5 L
t½	approximately 5 hours	k = 0.1386 hr^{-1} for $Cl = kV_D$

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = 12 \text{ mg/kg} \times 70 \text{ kg} = 840 \text{ mg}; \; V_D = 0.25 \text{ L/kg} \times 70 \text{ kg} = 17.5 \text{ L}$$

Dose and volume are both per kilogram, so each is multiplied by the 70-kg weight; the kilograms cancel and leave mg and L.

SETUP

$$k = \frac{0.693}{5 \text{ hr}} = 0.1386 \text{ hr}^{-1}; \; Cl_T = k \times V_D = 0.1386 \times 17.5 \text{ L} = 2.426 \text{ L/hr}$$

Clearance is not given, so it is built from the half-life and the volume: $Cl = kV_D$, total body clearance. hr^{-1} × L leaves L/hr.

ALGEBRA

$$AUC = \frac{F \, D_0}{Cl_T} = \frac{840 \text{ mg}}{2.426 \text{ L/hr}} = 346.3 \text{ (mg/L)·hr}$$

$Cl = \frac{F \, D_0}{AUC}$ rearranged. F = 1 for an IV bolus, because the whole dose goes directly into the blood. mg over L/hr leaves (mg/L)·hr.

ROUND

$$AUC = 346.3 \text{ (mg/L)·hr} = \frac{C_0}{k} = \frac{48 \text{ mg/L}}{0.1386 \text{ hr}^{-1}}$$

The AUC is the total exposure from this one dose, the quantity an exposure–response relationship sets against the effect.

> **DOES THE NUMBER MAKE SENSE?**
> $AUC = C_0 \div k$, with $C_0 = 840 \div 17.5 = 48$ mg/L and k = 0.1386 hr^{-1}: 346.3 (mg/L)·hr.

Her homework wording with different numbers, so this is not the graded answer.

1b

Worked by the drill

Answer: 17.93 hr

WHAT THE STEM GIVES

Ceff	4 mg/L	the MEC, where the duration ends

D_0	12 mg/kg	over 0.25 L/kg gives C_0 = 48 mg/L
V_D	0.25 L/kg	the denominator of C_0
t½	approximately 5 hours	k = 0.1386 hr^{-1}
weight	70-kg	cancels in C_0

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{D_0}{V_D} = \frac{12\ mg/kg}{0.25\ L/kg} = 48\ mg/L;\ k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$$

Dose and volume are both per kilogram, so the kilograms cancel in C_0: 840 mg over 17.5 L is the same 48 mg/L. k comes from the half-life.

ALGEBRA

$$teff = \frac{\ln C_0 - \ln Ceff}{k} = \frac{\ln(48 \div 4)}{0.1386\ hr^{-1}} = \frac{\ln 12}{0.1386} = \frac{2.485}{0.1386\ hr^{-1}}$$

The duration is the time the level takes to fall from C_0 to the MEC; the logarithm of their ratio over k gives that time.

ROUND

$$teff = 17.93\ hr$$

2.485 ÷ 0.1386 = 17.93 hr: between 3 and 4 half-lives of 5 hr, because the 12-fold fall lies between 8 and 16.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 ÷ Ceff = 12 lies between 8 and 16, so teff is between 3 and 4 half-lives of 5 hr: 17.93 hr.

Her homework wording with different numbers, so this is not the graded answer.

1c

Worked by the drill

Answer: 12.93 hr

WHAT THE STEM GIVES

D_0	6 mg/kg	half of 12 mg/kg, so C_0 halves to 24 mg/L
Ceff	4 mg/L	the MEC from part b
V_D	0.25 L/kg	unchanged by the dose
t½	approximately 5 hours	the amount teff falls by

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{6\ mg/kg}{0.25\ L/kg} = 24\ mg/L;\ k = 0.1386\ hr^{-1};\ Ceff = 4\ mg/L$$

Half the dose gives half of C_0, because the volume does not change with the dose; k and the MEC are the same as in part b.

ALGEBRA

$$\text{teff} = \frac{\ln(24 \div 4)}{0.1386\ \text{hr}^{-1}} = \frac{\ln 6}{0.1386} = \frac{1.792}{0.1386\ \text{hr}^{-1}} = 12.93\ \text{hr}$$

The level now has to fall only sixfold, from 24 to 4 mg/L, to reach the MEC, instead of twelvefold, so the duration is shorter than in part b.

ROUND

17.93 − 12.93 = 5 hr, one half-life; teff = 12.93 hr

Halving the dose removes one doubling, and each doubling adds one half-life, so 5 hr comes off the duration; the 17.93 hr is not halved.

DOES THE NUMBER MAKE SENSE?

Half the dose removes one half-life, 5 hr, from 17.93 hr: 12.93 hr, not half of 17.93.

Her homework wording with different numbers, so this is not the graded answer.

1d

Worked by the drill

Answer: 279.9 mg

WHAT THE STEM GIVES

teff	10 hours	the target duration, in the exponent
Ceff	4 mg/L	the MEC from part b
V_D	0.25 L/kg	× 70 kg = 17.5 L
t½	approximately 5 hours	k = 0.1386 hr^{-1}
weight	70-kg	scales the volume

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.25\ \text{L/kg} \times 70\ \text{kg} = 17.5\ \text{L};\ k = \frac{0.693}{5\ \text{hr}} = 0.1386\ \text{hr}^{-1}$$

The dose line wants the patient's volume in L, so the per-kilogram volume is multiplied by 70 kg; the half-life gives k for the exponent.

ALGEBRA

$$D_0 = \text{Ceff} \times V_D \times e^{k\ \text{teff}} = (4\ \text{mg/L})(17.5\ \text{L})\ e^{(0.1386)(10)} = 70\ \text{mg} \times e^{1.386}$$

$C = C_0 e^{-kt}$ with $C_0 = D_0$ over V_D, solved for the dose at the moment the level reaches the MEC. mg/L × L leaves 70 mg, the amount in the body at the MEC.

ALGEBRA

$e^{1.386} = 3.999$, so $D_0 = 70\ \text{mg} \times 3.999 = 279.9\ \text{mg}$

10 hours is two half-lives of 5 hr, so the level falls fourfold over that time and the dose must be four times the 70 mg at the MEC.

ROUND

$D_0 = 279.9$ mg (4.0 mg/kg for 70 kg)

279.9 mg, or 4.0 mg/kg in the per-kilogram form the stem doses in; a lower dose than the 12 mg/kg, because 10 hours is shorter than 17.93.

DOES THE NUMBER MAKE SENSE?

10 hours is two half-lives of 5 hr, so the dose is 4 times the 70 mg at the MEC: 279.9 mg.

Her homework wording with different numbers, so this is not the graded answer.

2a Worked by the drill

Answer: 5.21 mg/L

WHAT THE STEM GIVES

R1	250 mg	the first daily dose
C_1	6.4 mg/L	steady state on 250 mg
R2	350 mg	the increased daily dose
C_2	17.6 mg/L	steady state on 350 mg
weight	72-kg	*not needed for KM*

THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 250 mg/day, C_1 = 6.4 mg/L, R2 = 350 mg/day, C_2 = 17.6 mg/L

Two daily doses in the same patient and the steady-state concentration each produced: the direct method for KM.

ALGEBRA

$$KM = \frac{350 - 250}{(250/6.4) - (350/17.6)} = \frac{100}{39.063 - 19.886} = \frac{100}{19.177}$$

Each dose over its own concentration, carried to three decimals because the two ratios are close and rounding them early moves KM noticeably.

ROUND

KM = 5.21 mg/L

(mg/day) over (mg/day per mg/L) leaves mg/L, a concentration. The unrounded 5.215 mg/L is carried into Vmax and the later parts.

DOES THE NUMBER MAKE SENSE?

Raising the dose from 250 to 350 mg nearly tripled the concentration, so the drug is near saturation; KM is a concentration: 5.21 mg/L.

Her homework wording with different numbers, so this is not the graded answer.

2b Worked by the drill

Answer: 453.7 mg/day

WHAT THE STEM GIVES

KM	5.215 mg/L	from part a
R2	350 mg	the dose used
C_2	17.6 mg/L	its own steady state

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax: multiply both sides by (KM + C_{ss}) and divide by C_{ss}, using one dose and the steady-state concentration it produced.

ALGEBRA

$$\text{Vmax} = \frac{(350\ \text{mg/day})(5.215 + 17.6)\ \text{mg/L}}{17.6\ \text{mg/L}} = \frac{7985.25}{17.6} = 453.7\ \text{mg/day}$$

mg/L cancels and Vmax keeps the units of the dose, mg/day. The 250 mg/day with 6.4 mg/L pair gives 453.7 mg/day as well.

ROUND

$$\text{Vmax} = 453.7\ \text{mg/day}$$

453.71 mg/day to one decimal place. It is above the 350 mg/day dose, as it must be for a dose that reached a steady state.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax must exceed the 350 mg/day dose that reached a steady state: 453.7 mg/day.

Her homework wording with different numbers, so this is not the graded answer.

2c Worked by the drill

Answer: 330 mg/day

WHAT THE STEM GIVES

KM	5.215 mg/L	from part a
Vmax	453.7 mg/day	from part b
C_{ss}	14 mg/L	the target
rounding	nearest ten	330.6 becomes 330

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{\text{Vmax} \times C_{ss}}{KM + C_{ss}}$$

The patient's own KM and Vmax, from parts a and b, with the target of 14 mg/L, in the nonlinear dose-rate line.

ALGEBRA

$$R = \frac{(453.7\ \text{mg/day})(14\ \text{mg/L})}{(5.215 + 14)\ \text{mg/L}} = \frac{6351.8}{19.215} = 330.6\ \text{mg/day}$$

mg/L cancels and leaves mg/day. 14 mg/L lies between 6.4 and 17.6 mg/L, so the dose lies between 250 and 350 mg/day.

ROUND

$R = 330$ mg/day

The stem asks for the nearest ten, so 330.6 mg/day is reported as 330 mg/day, a dose that can be given and that keeps C_{ss} near 14 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> 14 mg/L lies between 6.4 and 17.6 mg/L, so the dose lies between 250 and 350 mg/day: 330 mg/day.

Her homework wording with different numbers, so this is not the graded answer.

2d

Worked by the drill

Answer: 1.48 days

WHAT THE STEM GIVES

weight	72-kg	$V_D = 0.7 \times 72 = 50.4$ L
V_D	70% of body weight	0.7 L/kg
C_p	14 mg/L	the concentration the dose achieves
KM	5.215 mg/L	from part a
Vmax	453.7 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

UNIT

$V_D = 0.70 \times 72$ kg $= 50.4$ L

"70% of body weight" is read as 0.7 L/kg, taking 1 L as 1 kg, so the volume is 0.7 times the 72 kg weight, in litres.

SETUP

$t½ = V_D(\frac{0.693}{Vmax})(KM + C_p)$, at $C_p = 14$ mg/L

The recommended dose was chosen to give 14 mg/L, so the half-life is calculated at that concentration; at another concentration it would differ.

ALGEBRA

$$t½ = 50.4 \text{ L} \times \frac{0.693}{453.7 \text{ mg/day}} \times (5.215 + 14) \text{ mg/L} = 1.479 \text{ days}$$

L × mg/L gives mg, which cancels against mg/day and leaves days. Using the volume as 0.7 gives 0.02, a result per kilogram.

ROUND

$t½ = 1.48$ days

1.479 days to two decimal places. The rounded 330 mg/day dose gives 13.9 mg/L and 1.47 days, within the same tolerance.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax is in mg/day, so the half-life comes out in days: 1.479 days, more than one day and less than two.

Her homework wording with different numbers, so this is not the graded answer.

2d

Worked by the drill

Answer: 23.61 L/day

WHAT THE STEM GIVES

C_p	14 mg/L	added to KM in the denominator
KM	5.215 mg/L	from part a
Vmax	453.7 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}, \text{ at } C_p = 14 \text{ mg/L}$$

The recommended dose was chosen to give 14 mg/L, and the clearance of a saturable drug is calculated at its concentration.

ALGEBRA

$$Cl_T = \frac{453.7 \text{ mg/day}}{(5.215 + 14) \text{ mg/L}} = \frac{453.7}{19.215} = 23.61 \text{ L/day}$$

mg/day over mg/L leaves L/day, a volume of plasma cleared per unit time. The volume of distribution is not needed for clearance.

ROUND

$$Cl_T = 23.61 \text{ L/day}$$

To two decimal places. A higher dose would raise C_p and lower this clearance, because C_p is in the denominator.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance at 14 mg/L is well below Vmax ÷ KM, because KM + C_p, 19.215 mg/L, is larger than KM alone: 23.61 L/day.

Her homework wording with different numbers, so this is not the graded answer.

2e

Worked by the drill

Answer: 10.18 mg/L

WHAT THE STEM GIVES

R	300 mg/day	the new dose rate
KM	5.215 mg/L	from part a
Vmax	453.7 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R \times KM}{Vmax - R}$$

The dose-rate line solved for C_{ss}: multiplying out gives $R \times KM + R \times C_{ss} = Vmax \times C_{ss}$, so $R \times KM = C_{ss}(Vmax - R)$, then divide by $(Vmax - R)$.

ALGEBRA

$$C_{ss} = \frac{(300 \text{ mg/day})(5.215 \text{ mg/L})}{(453.7 - 300) \text{ mg/day}} = \frac{1564.5}{153.7} = 10.18 \text{ mg/L}$$

mg/day cancels and leaves mg/L. The denominator is Vmax minus the dose; adding them gives 2.1 mg/L, below the 6.4 mg/L that 250 mg produced.

ROUND

$$C_{ss} = 10.18 \text{ mg/L}$$

10.18 mg/L to two decimal places. 300 mg/day is 20% above 250 mg/day, yet C_{ss} rises 59% from 6.4 mg/L: the steep rise of a saturable drug.

DOES THE NUMBER MAKE SENSE?

300 mg/day is above the 250 mg/day that gave 6.4 mg/L and below the dose that gives 14 mg/L, so C_{ss} lies between them: 10.18 mg/L.

Her homework wording with different numbers, so this is not the graded answer.